

Experiment No. 3

DETERMINATION OF THE ANGULAR DIVERGENCE OF THE LASER BEAM

Aim: To determine the angular divergence of the laser beam.

Apparatus: 2 mW He Ne Laser, Optical bench/table and a graph sheet as a Screen.

Formula: Half of the Angular divergence, $\theta_0 = \frac{R_2 - R_1}{D_2 - D_1}$ radians

Where, R_1 is the spot radius at position-1 in cm.

R_2 is the spot radius at position-2 in cm.

D is the distance of the screen in cm measured from the laser position.

THEORY:

When a laser beam is allowed to fall on a screen, a circular bright spot is seen on the screen. If screen is moved near or far from the source, the size of the spot decreases or increases respectively.

Note that the beam will have an initial beam diameter and that diameter will increase over distance. This increase is known as beam divergence. The approximate beam divergence in radians can be obtained.

The light emitted by a laser is confined to a narrow cone. But, when the beam propagates outward, it slowly diverges or fans out. For an electromagnetic beam, beam divergence is the angular measure of the increase in the radius or diameter with distance from the optical aperture as the beam emerges.

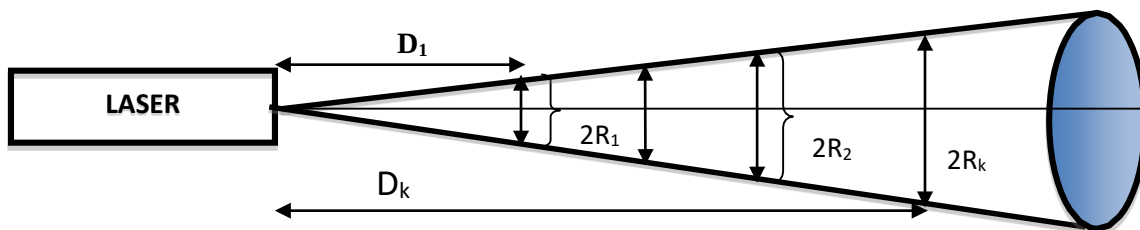
The divergence of a laser beam can be calculated if the beam diameter ω_1 ($2R_1$) and ω_2 ($2R_2$) at two separate distances are known. Let D_1 and D_2 be the distances along the laser axis, from the end of the laser to position “1” and “2”.

PROCEDURE:

1. Laser tube is mounted on a tube holder.

2. A screen is placed at about 50 cm from the laser.
3. The laser is switched on. It forms a red spot on the screen.
4. The diameter of the spot $2R_1$ on the screen is measured.
5. The screen is moved around 100 cm away. The diameter of the spot $2R_2$ is measured which is the second position.
6. The distance $D = (D_2 - D_1)$ is measured between the two positions of the screen.
7. D should not be less than 50 cm. It should be preferable around 100 cm
8. The angular divergence θ_0 , is calculated from the formula given above.
9. Angular divergence is calculated for different screen position using the above mentioned procedure and mean angular divergence is calculated.
10. Now plot the beam radius and the corresponding position on a graph. The slope of this graph gives the divergence.
11. Compare the two values obtained.

DIAGRAM 1:



OBSERVATIONS:

Sr. No	Position (D_k) (cm)	R_k (cm)	$\theta_0 = \frac{R_m - R_n}{D_m - D_n}$	Mean θ_0 (rad)
1	50			
2	150			
3	250			
4	350			
5	450			
6	550			
7	650			
8	750			
9	850			

Calculation:

Graphical Analysis: Plot a graph between position (D) on x-axis and radius (R) on y-axis and find the slope.

Compare the calculated values of θ from above two methods

Error analysis: $\theta = \frac{\Delta R}{\Delta D}$; we can estimate the error as

$$\% \text{ error in } \theta = \% \text{ error in } \Delta R + \% \text{ error in } \Delta D$$

RESULT: The angular divergence of the He- Ne laser beam is found to be _____ $\pm$ _____ rad.

Precaution:

- (1) Spot size should be measured accurately.
- (2) Laser light should not fall directly to the eyes of the observer.
- (3) Spot size of laser should be circular in shape.